Graphic User Interface for easy nuclear reaction calculation

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Introduction

- A development of Graphic User Interface for nuclear reaction calculation
- For novice and experts who wants a quick calculation.

- Actually, this is not a new work.
  - Presentated at KPS

- Motivation of presentation again?
  - Updates in the GUI since last time
  - Force myself to work.

Introduction

- There are many direct nuclear reaction codes available
  - DWUCK, ECIS, FRESCO, TWOFNR, CCFULL, PIKOE…
- Most of them are written in Fortran
  - Require specific input file format
  - Require knowledge on the reaction theory
  - Usually they are not easy for beginner to use
  - Usually they need separate plotting tools

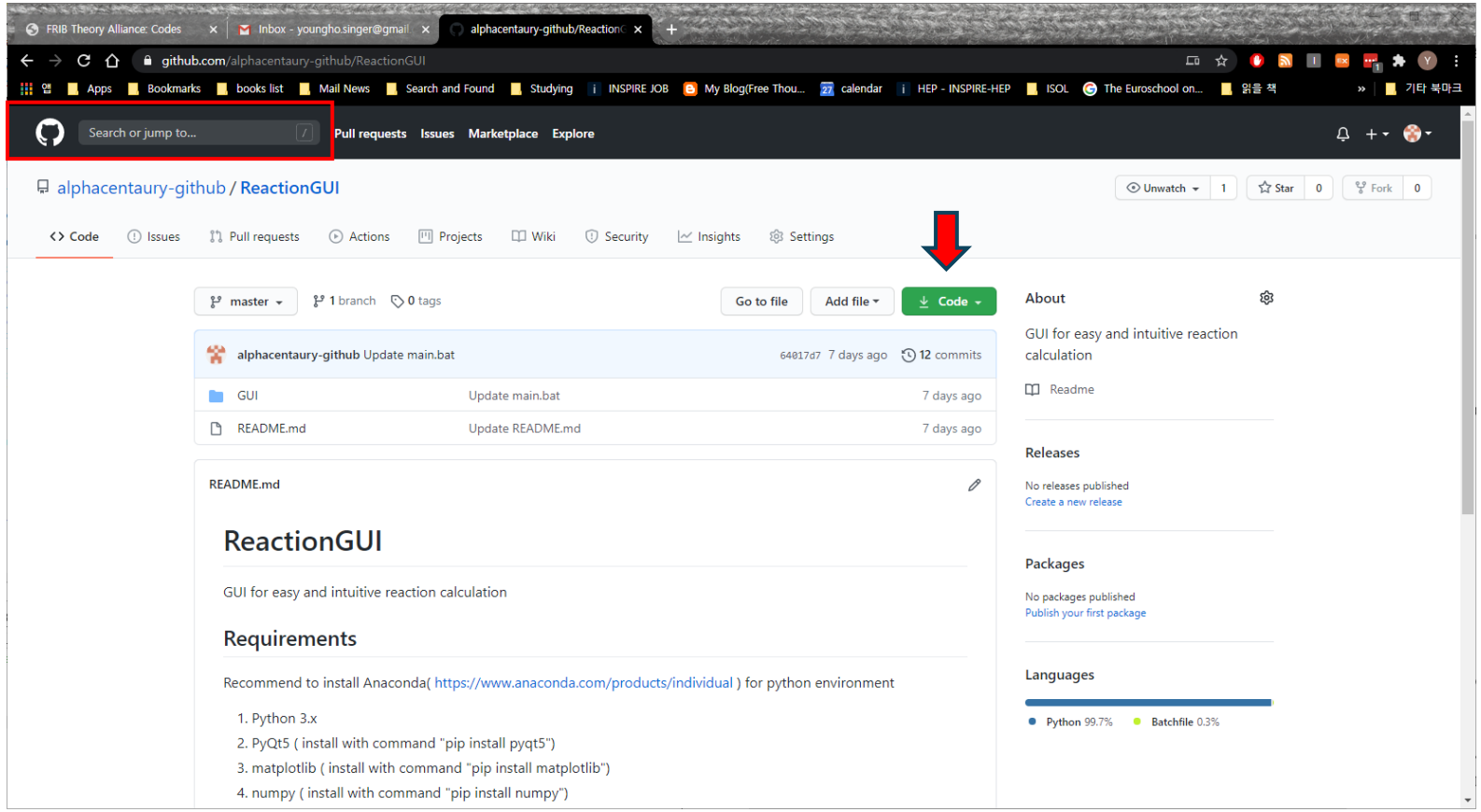
- The aim of the GUI is to make it easy and intuitive to do a nuclear reaction.

- However, there are large room for improvements.

Introduction

- GUI code is developed by Y.-H. Song and I.J. Shin.
- Combines Qt + Fresco + other utility codes
- Plan to improve the code according to user's needs.
  (yhsong@ibs.re.kr for suggestions, errors, troubles)
- Search "ReactionGUI" at GitHub
- https://github.com/alphacentaury-github/ReactionGUI

Introduction



Introduction

Requirements

Recommend to install Anaconda(https://www.anaconda.com/products/individual) for python environment

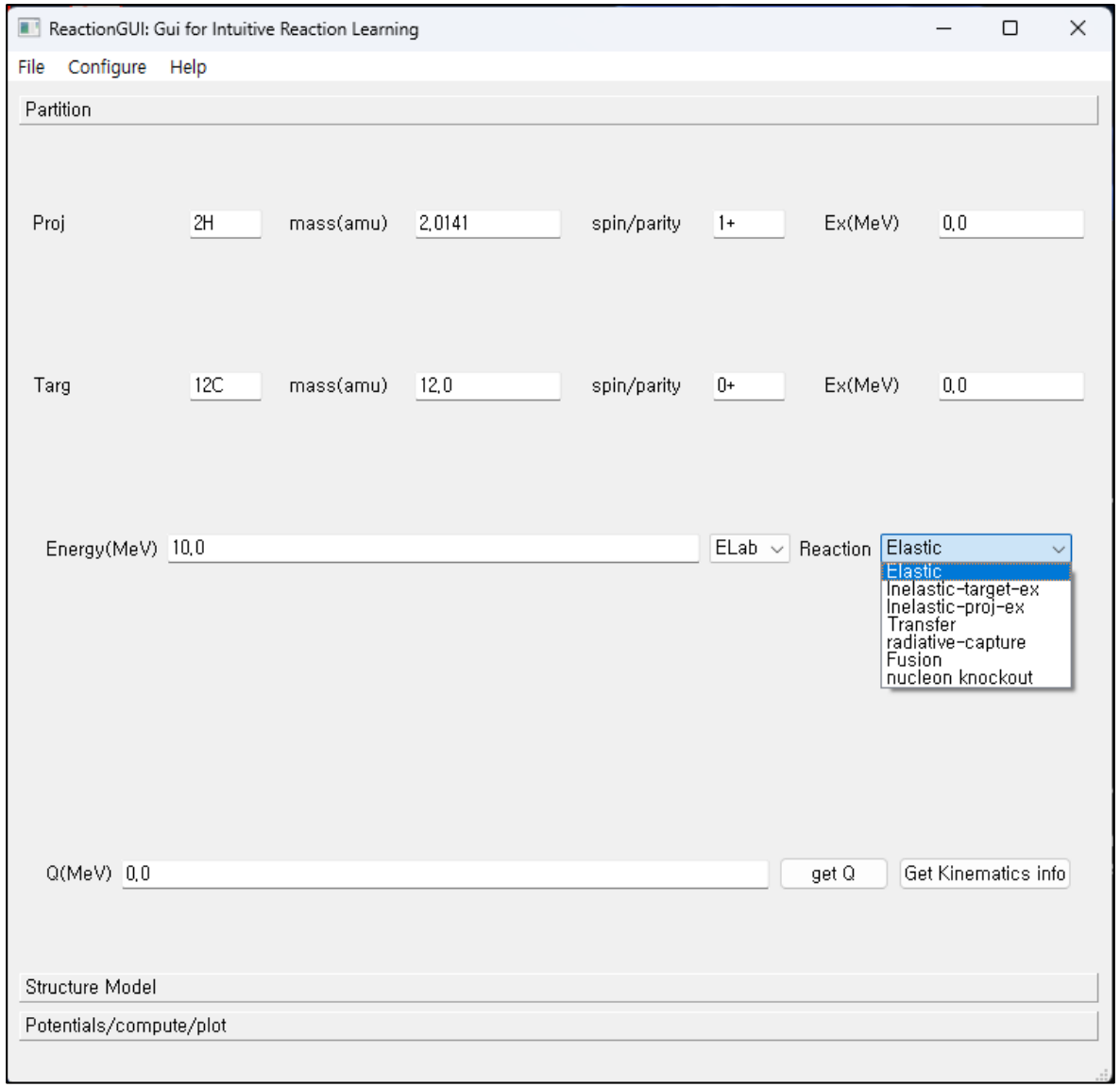
1. Python 3.8 or higher
2. PyQt5 (install with command "pip install pyqt5")
3. matplotlib (install with command "pip install matplotlib")
4. numpy (install with command "pip install numpy")
5. scipy (install with command "pip install scipy")

---followings are included and thus no separate install/download is not necessary. ---

6. executable file of FRESCO and SFRESCO. (from http://www.fresco.org.uk/)
7. executable file of OMGET and required data files. (from https://www-nds.iaea.org/RIPL-3/ (in the OPTICAL tab))
8. density profile data files from HFB-14 calculation. (from https://www-nds.iaea.org/RIPL-3/ (in the MASSES tab))

To run GUI: in command prompt
python main.py

GUI

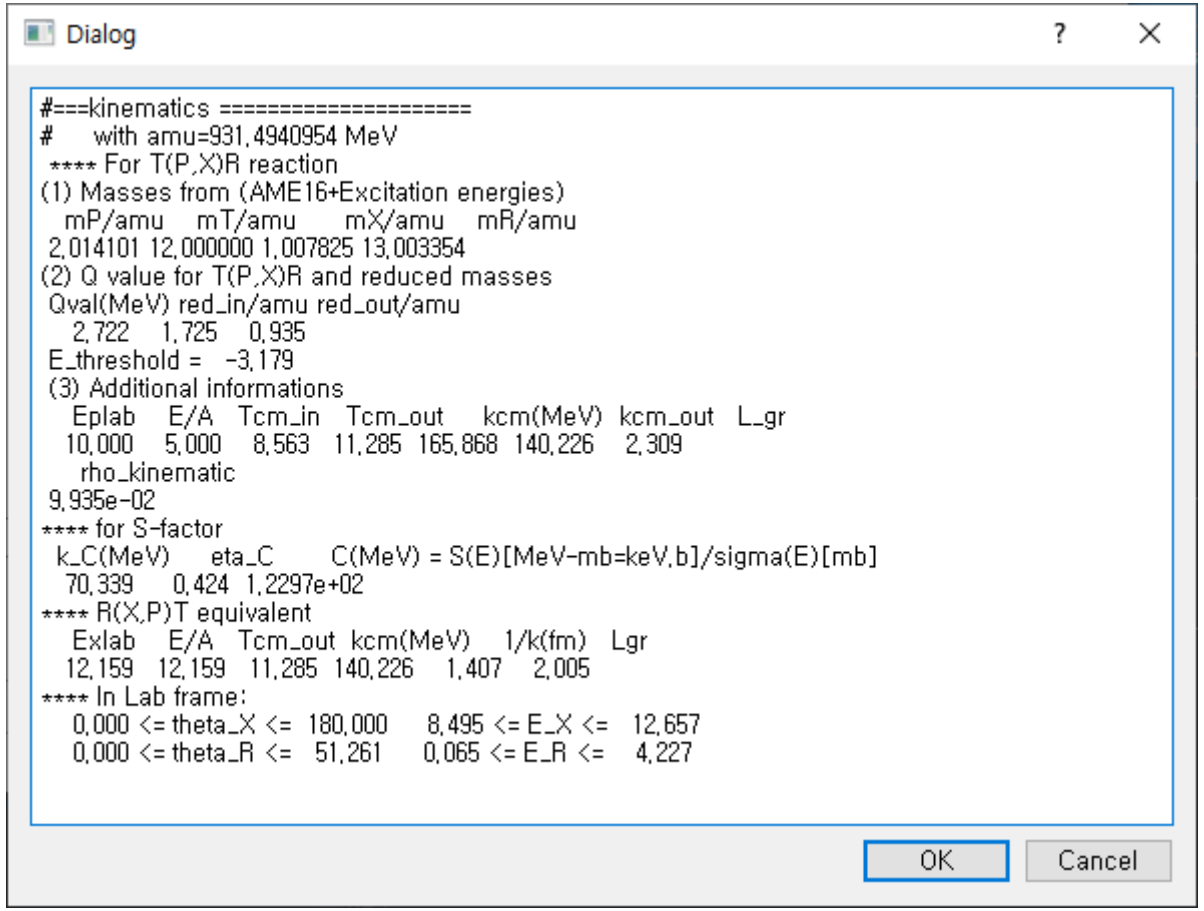


File:
Save/Load
Input parameter values

Configure:
Set executables (fresco, sfresco)

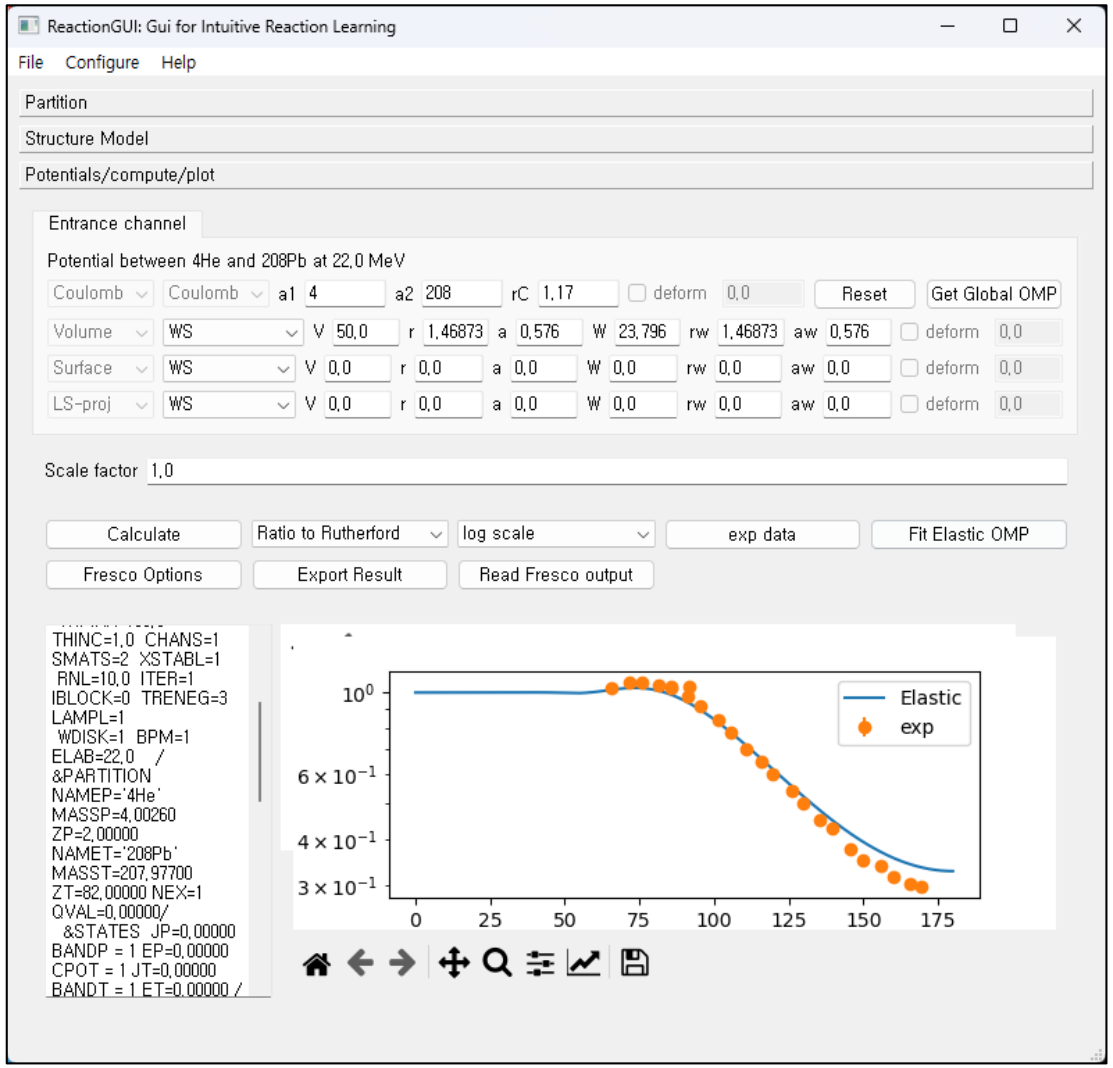
Three tabs
1. Partition
2. Structure Model
3. Potentials/compute/plot

GUI

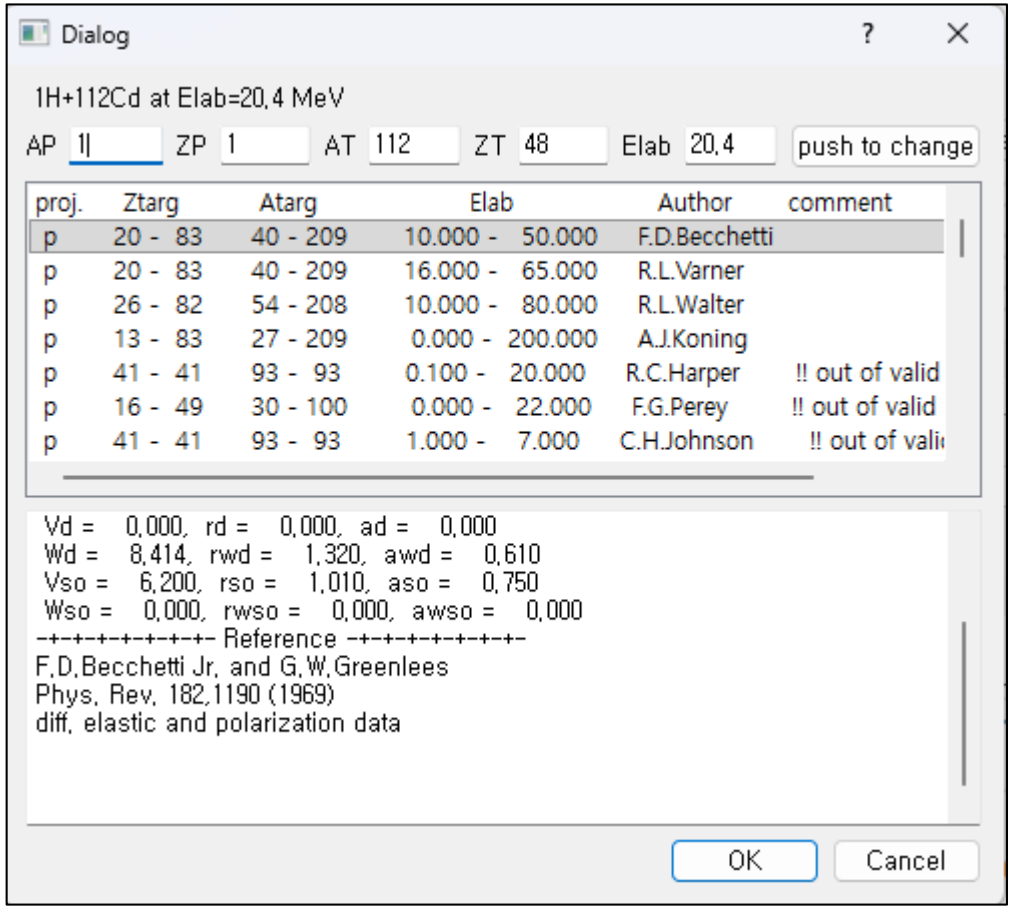


Get kinematics button

Elastic scattering: Optical model

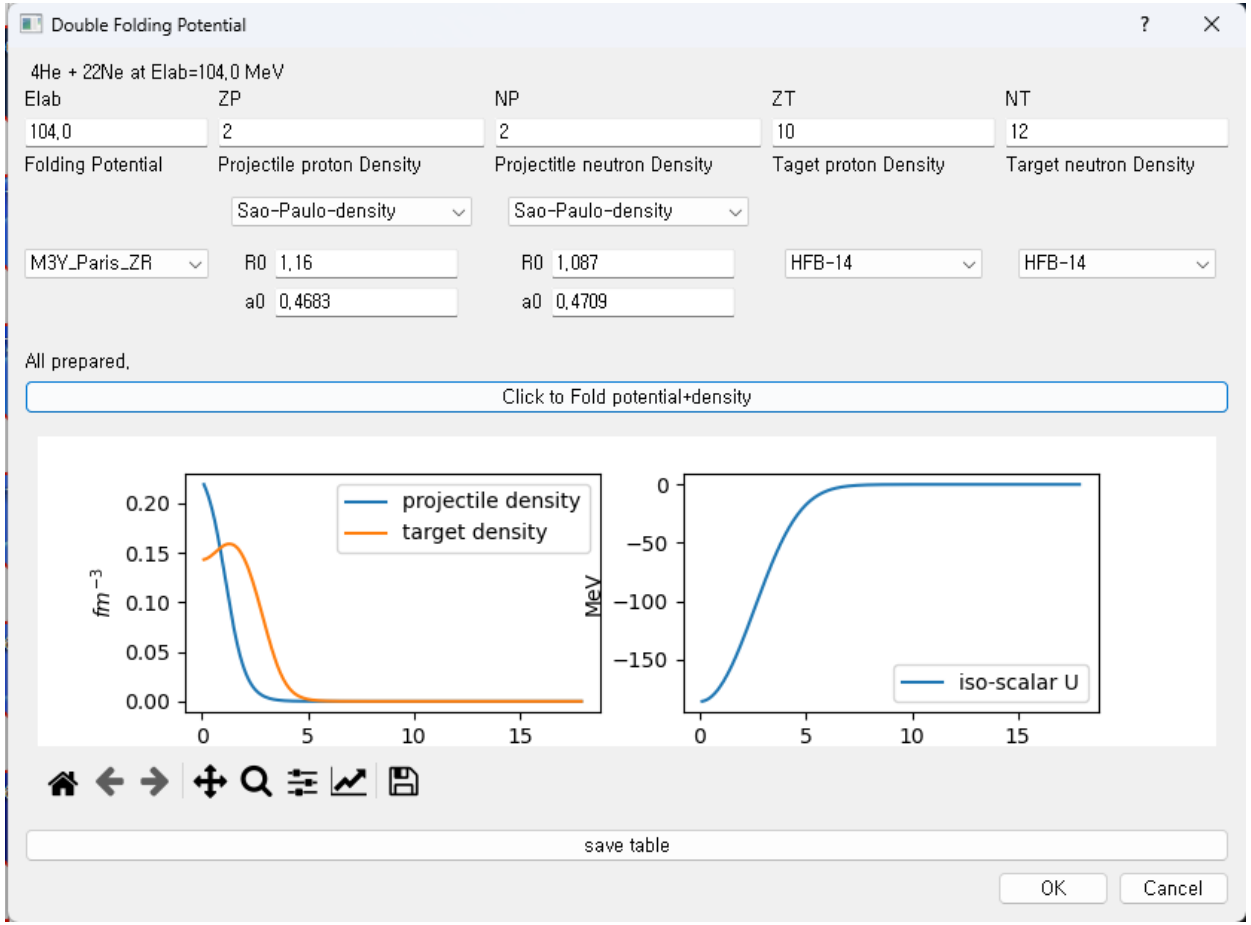


Elastic scattering: Optical model



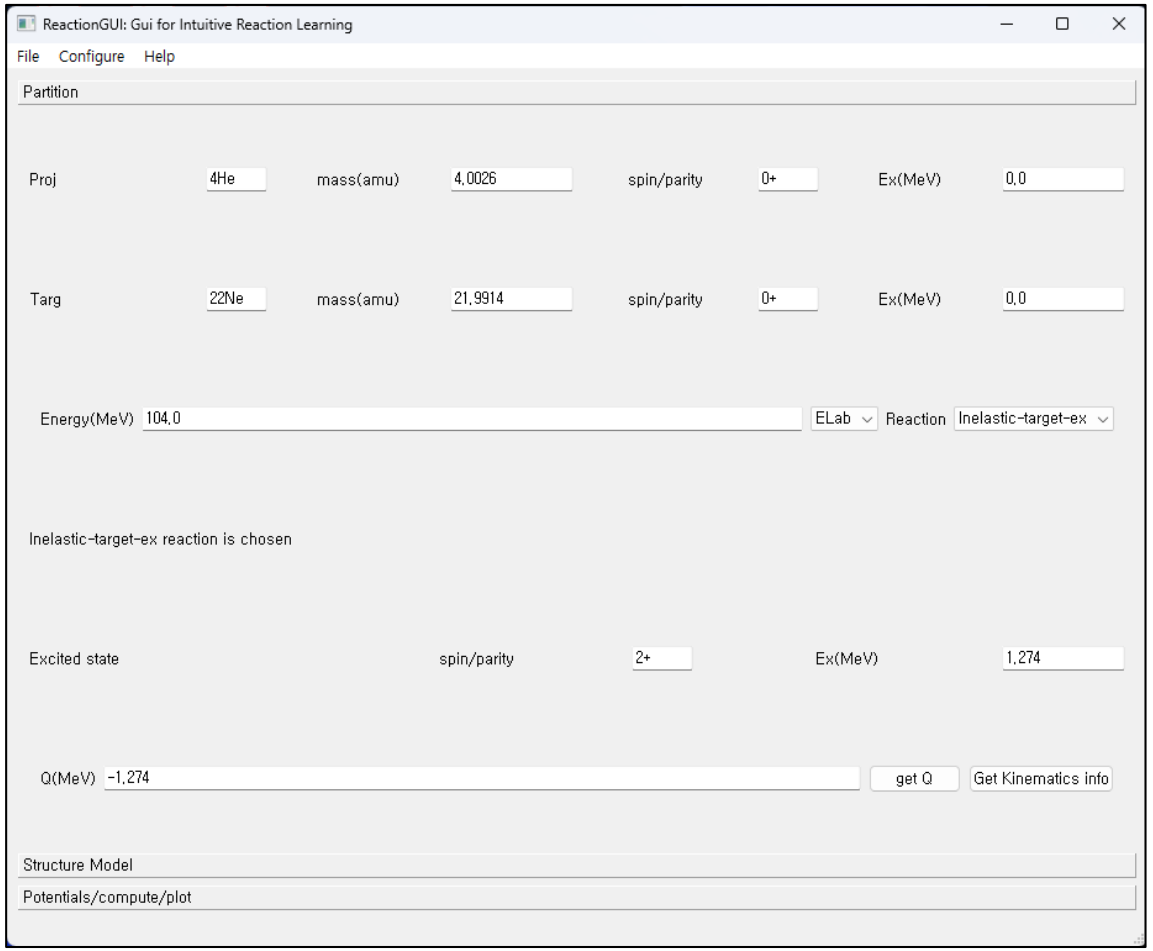
OMGET: get global optical potential parameters

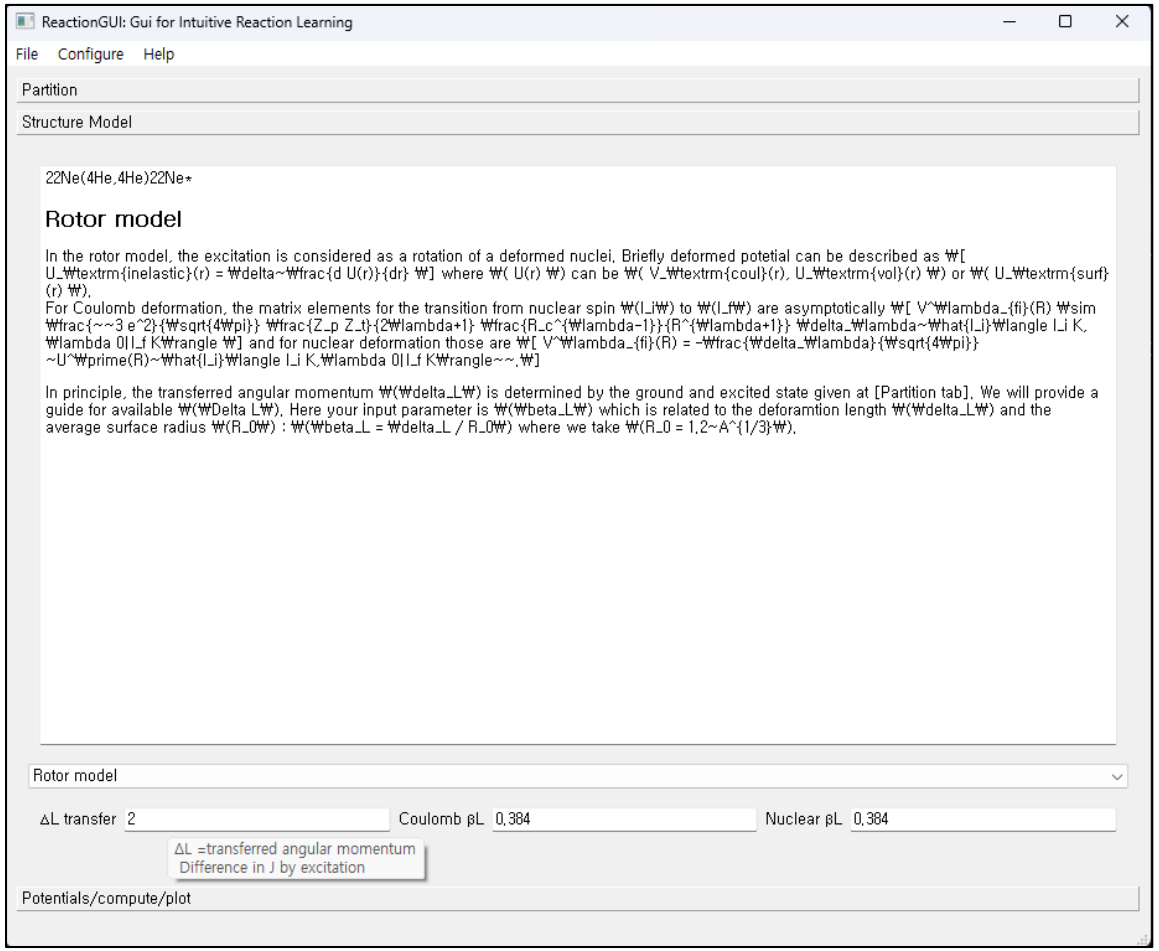
Elastic scattering: Optical model



Double Folding

Inelastic scattering: DWBA(Rigid Rotor model)





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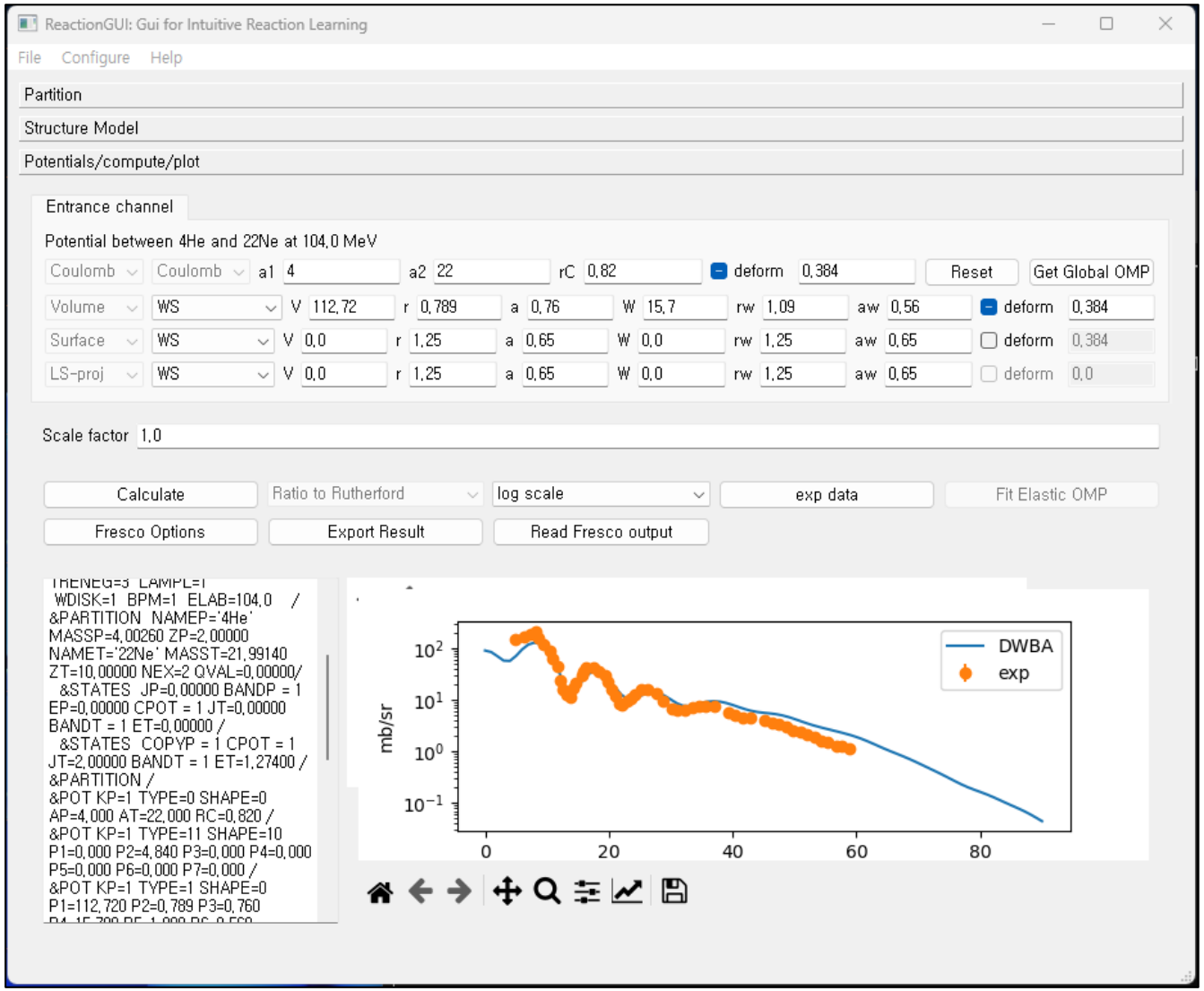
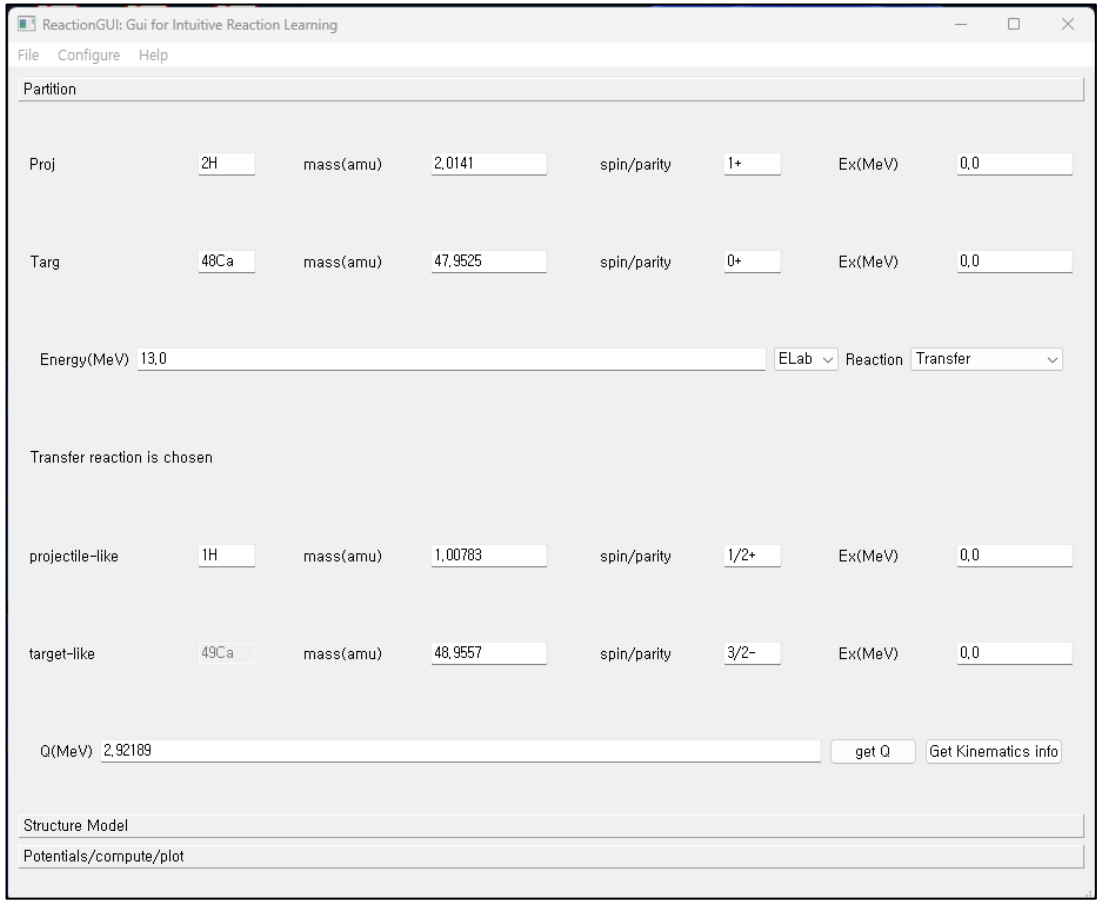
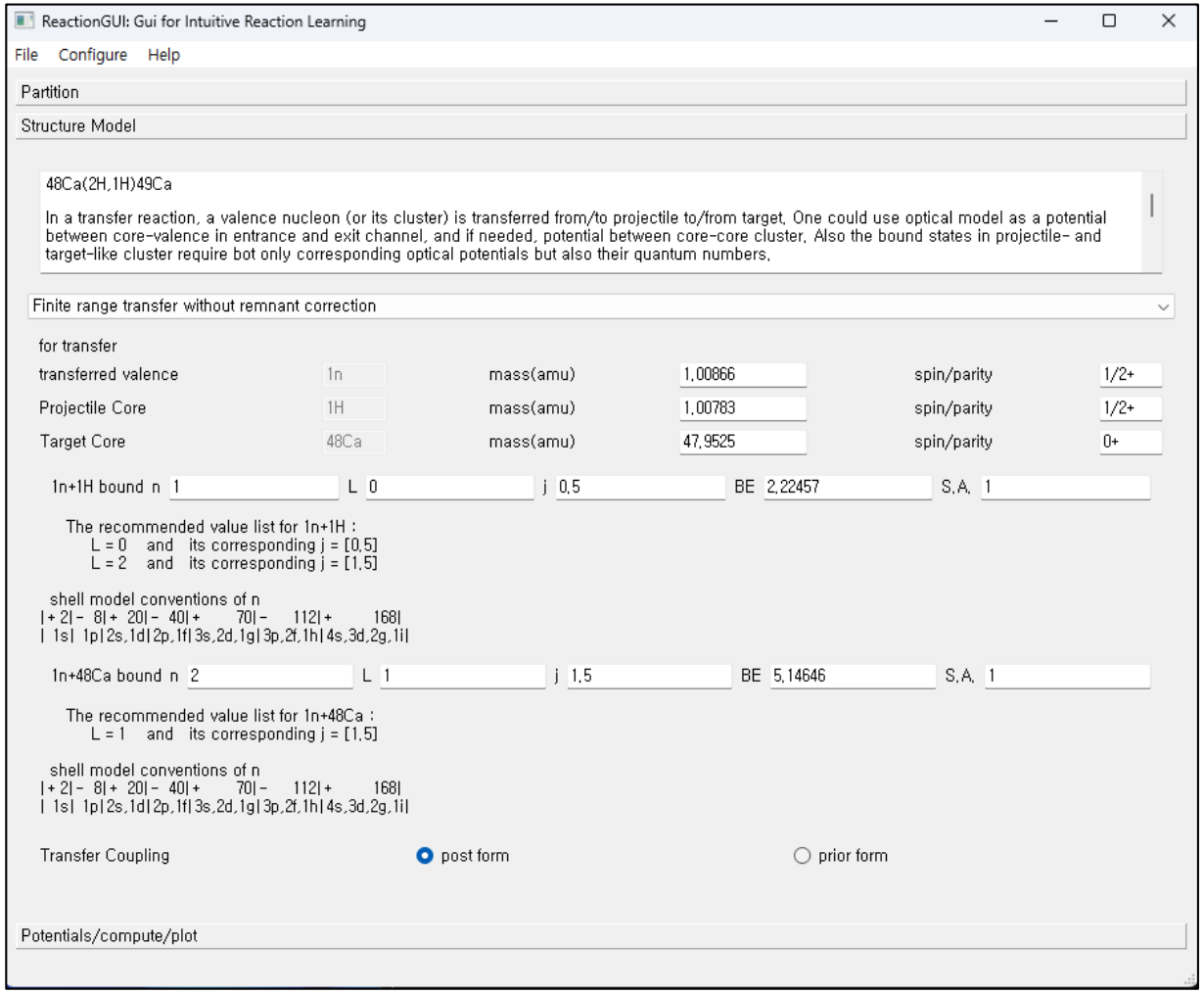


Transfer reaction: DWBA





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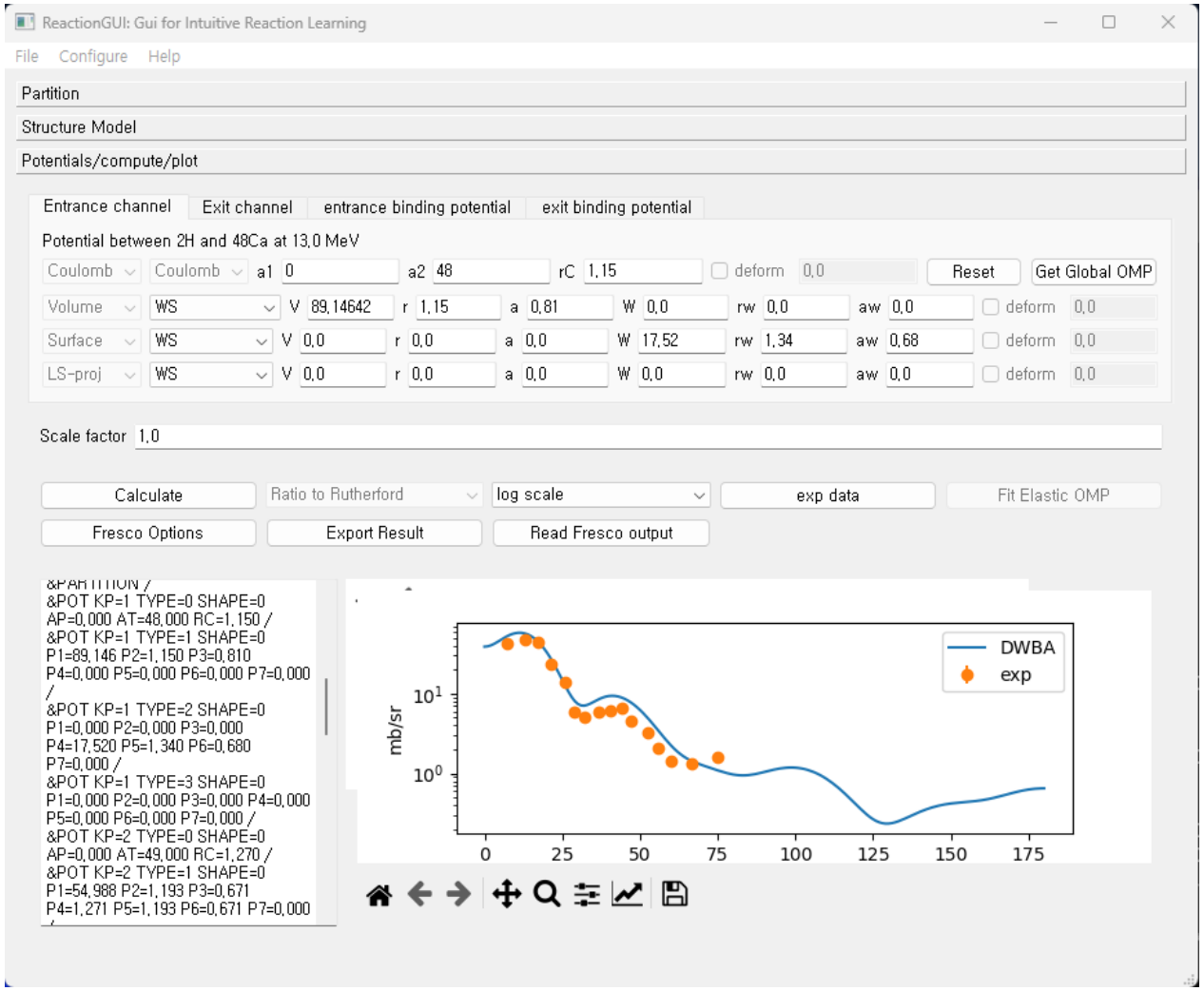
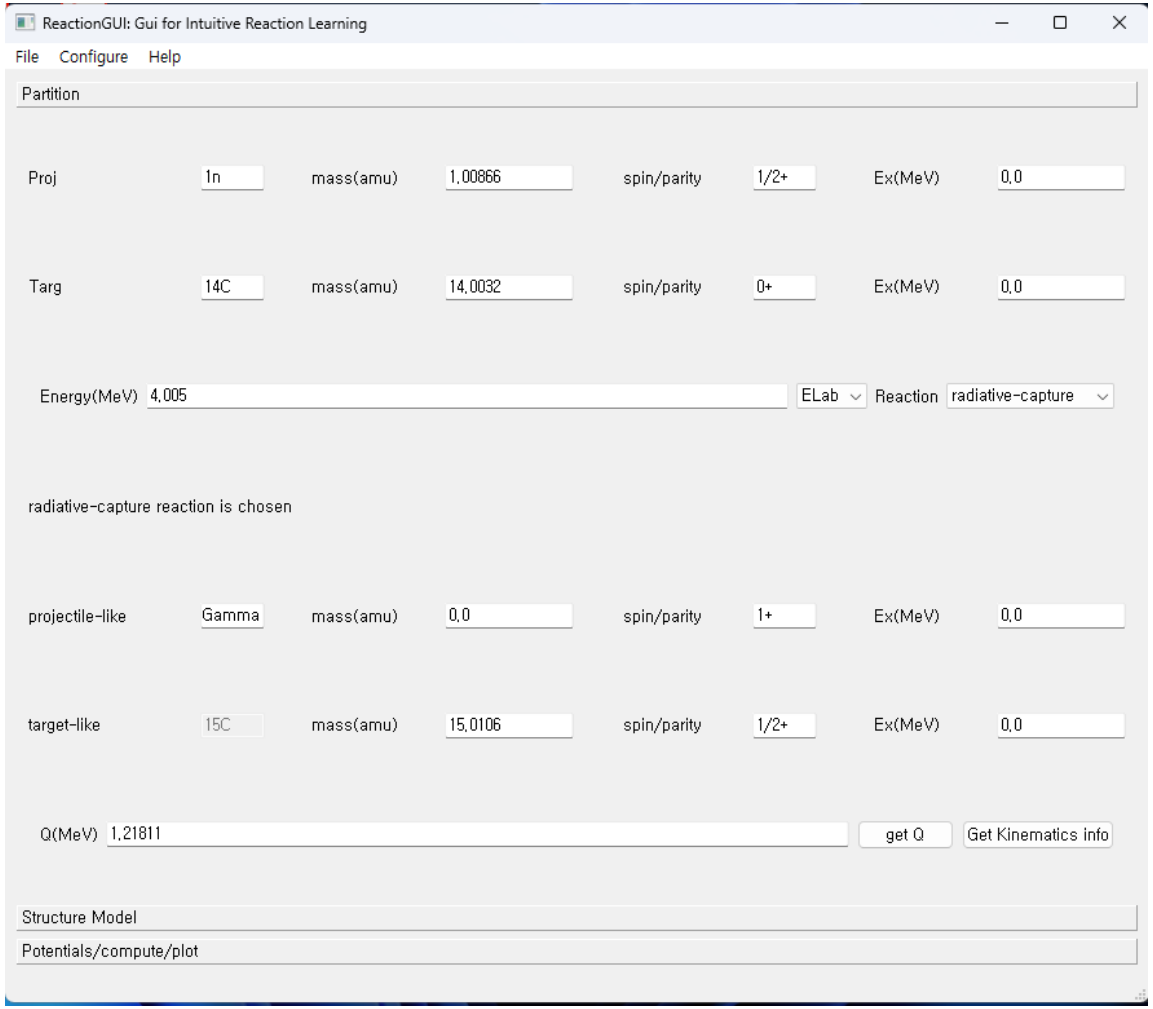
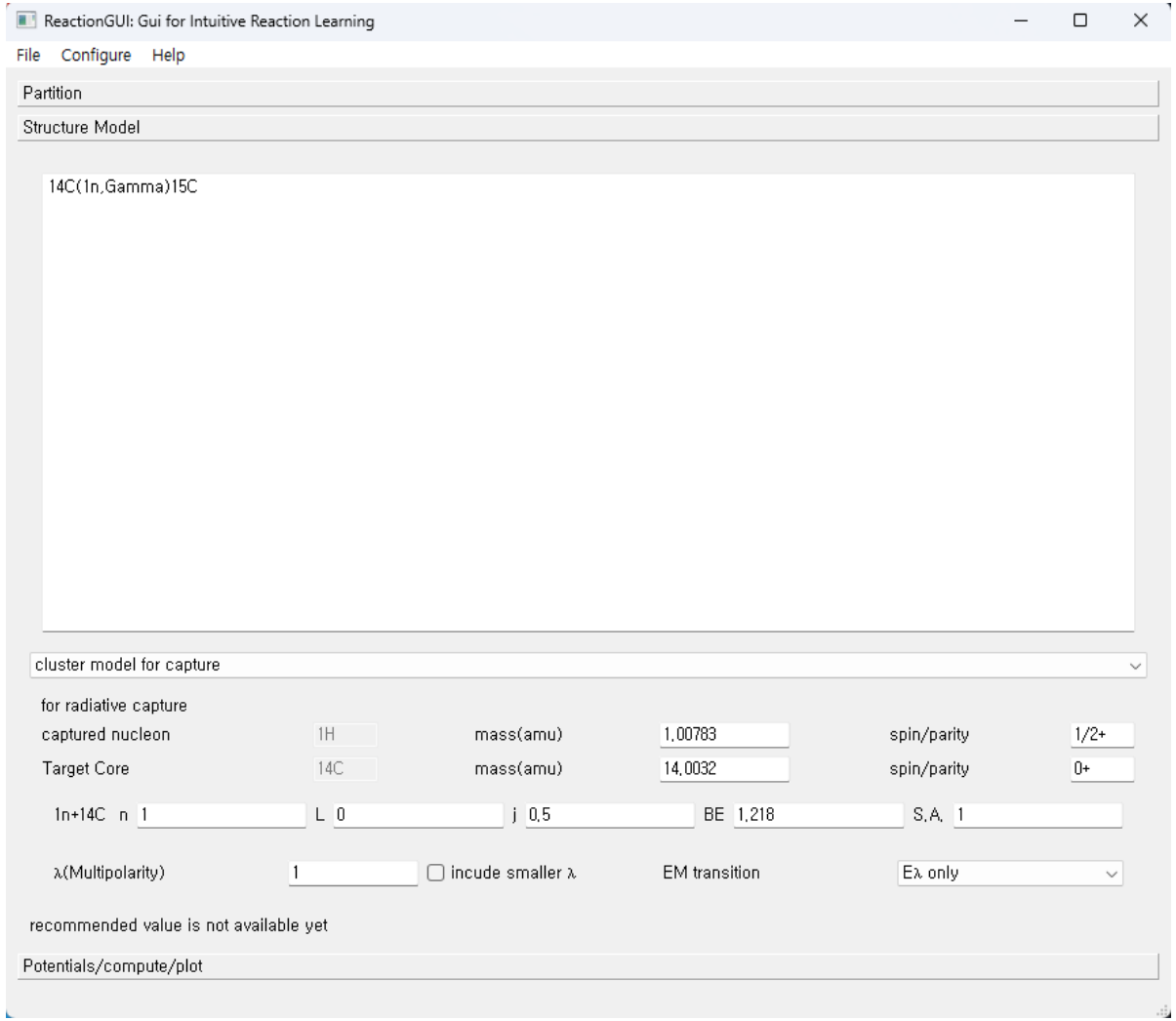


Radiative capture reaction: DWBA





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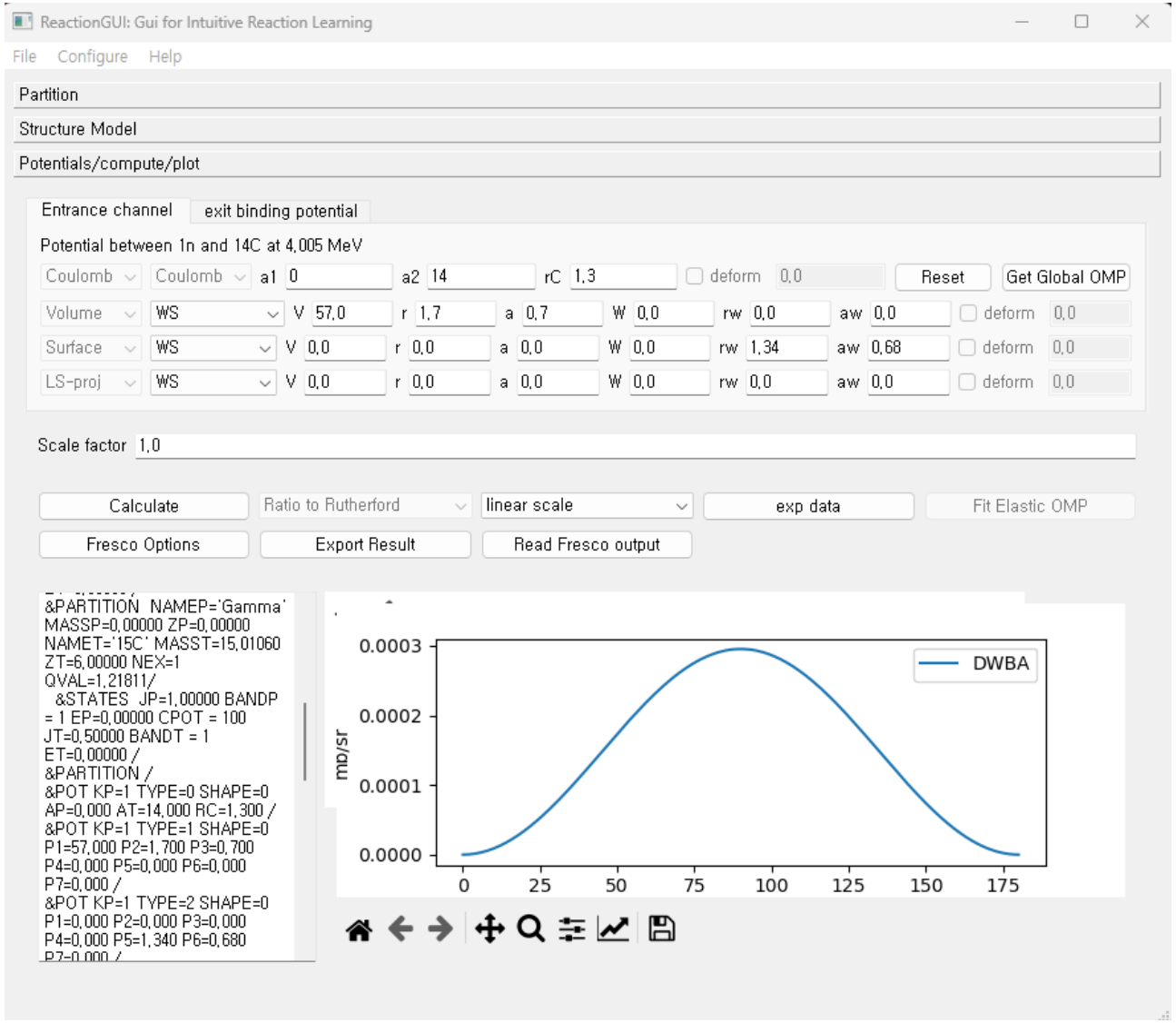

Works to do

- Fusion reaction:
  - Fusion cross section in optical model is available in elastic scattering.
  - GUI of coupled channel fusion reaction(CCFULL) is included, but as a separate window

- Knock-out reaction:
  - GUI of MOMDIS is included, but as a separate window
  - GUI of PIKOE is under development, but as a separate program.

- → Need to improve the interface and integrate all.

- Need to generalize GUI for coupled channel calculations.

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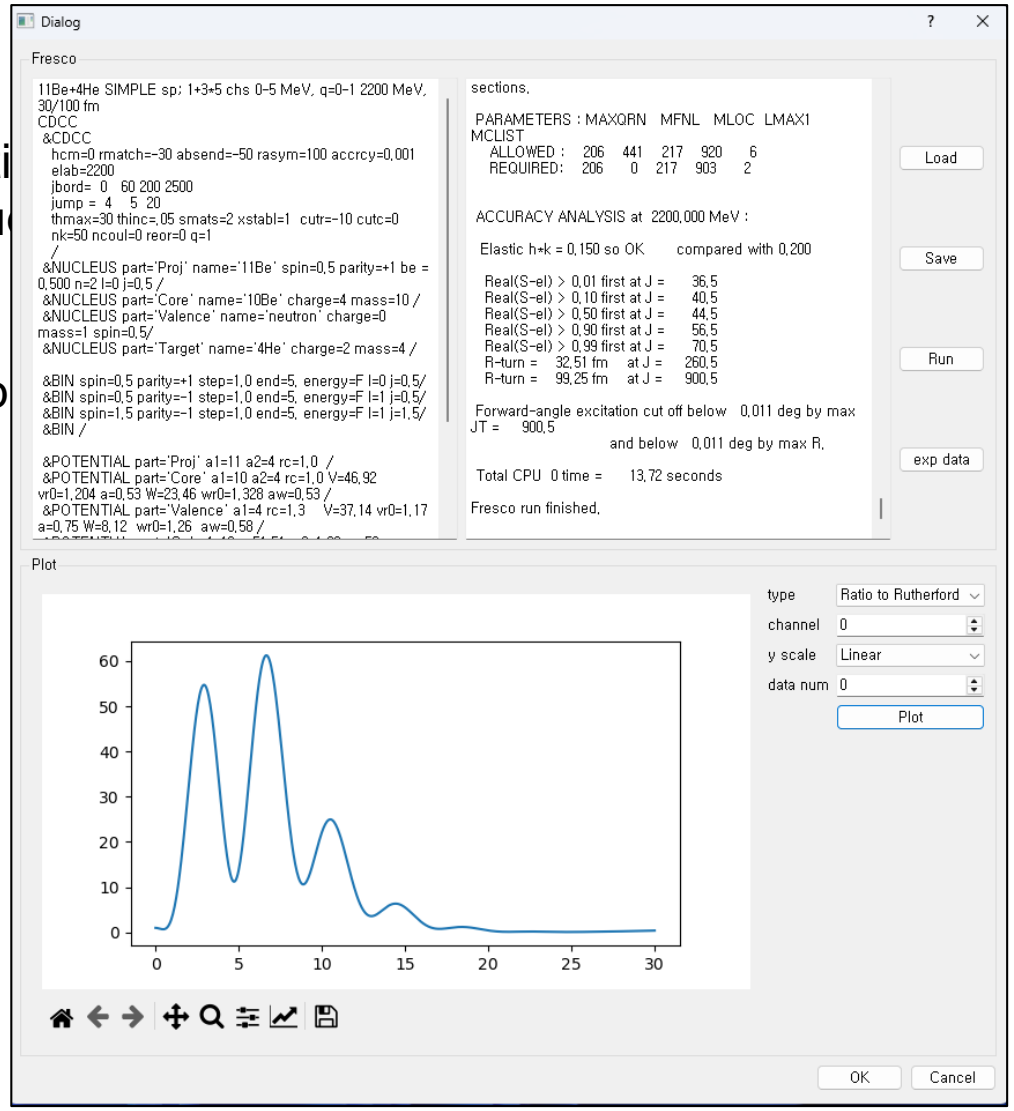
- For the moment, “qt_fresco.py” can be used
  - Edit the fresco input file
  - Run the code
  - Plot the output